

Cornell Notes: Data Mining Overview

Cue / Keywords	Notes
Data Mining	- Analytical process of sorting large datasets to find patterns. - Objective: Solve business problems using structured data analysis.
Industry	- Telecom/Media: Targeted ads, customer behavior prediction. - Insurance: Fraud detection, risk management, compliance. - Finance: Market risk evaluation, return maximization. - Retail: Product association (joint purchasing), optimized placement. - Healthcare: Accurate diagnoses via patient history and treatment patterns. - Manufacturing: Just-in-time fulfillment, predicting equipment failure.
Common Techniques	- Classification: Assigning data points into categories. - Clustering: Grouping based on shared characteristics. - Regression: Analyzing relationships between variables. - Prediction: Finding patterns/trends to calculate probability of future outcomes. - Association Rules: Discovering correlations between data points.

Summary

Data mining is the process of discovering patterns in large datasets to solve business problems. It has broad applications across various industries (telecom, finance, healthcare, etc.) to optimize processes and predict outcomes. Key techniques include classification, clustering, regression, prediction, and association rules.